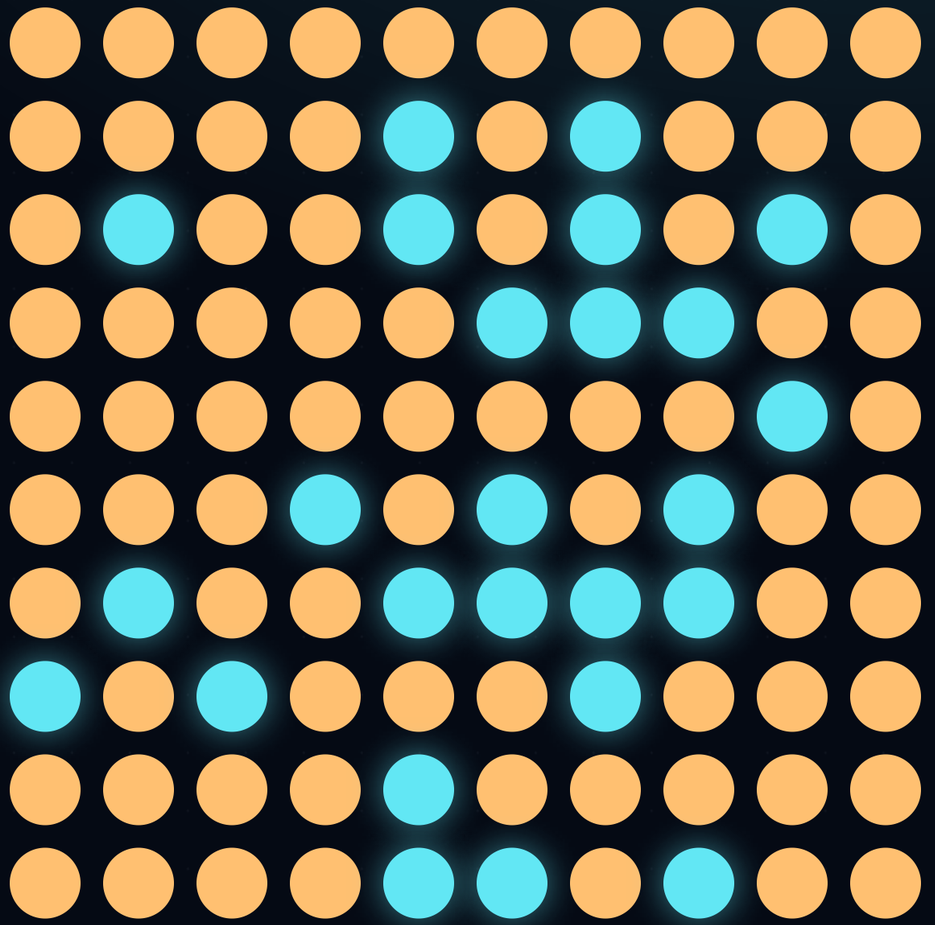
PH4040 THE TALK · OCTOBER 2026

Turning *dark* excitons into *light*

Thermally activated delayed fluorescence in organic semiconductors

- quantum spin
- molecular design
- OLED efficiency



- 25 singlets
- 75 triplets, recycled by TADF

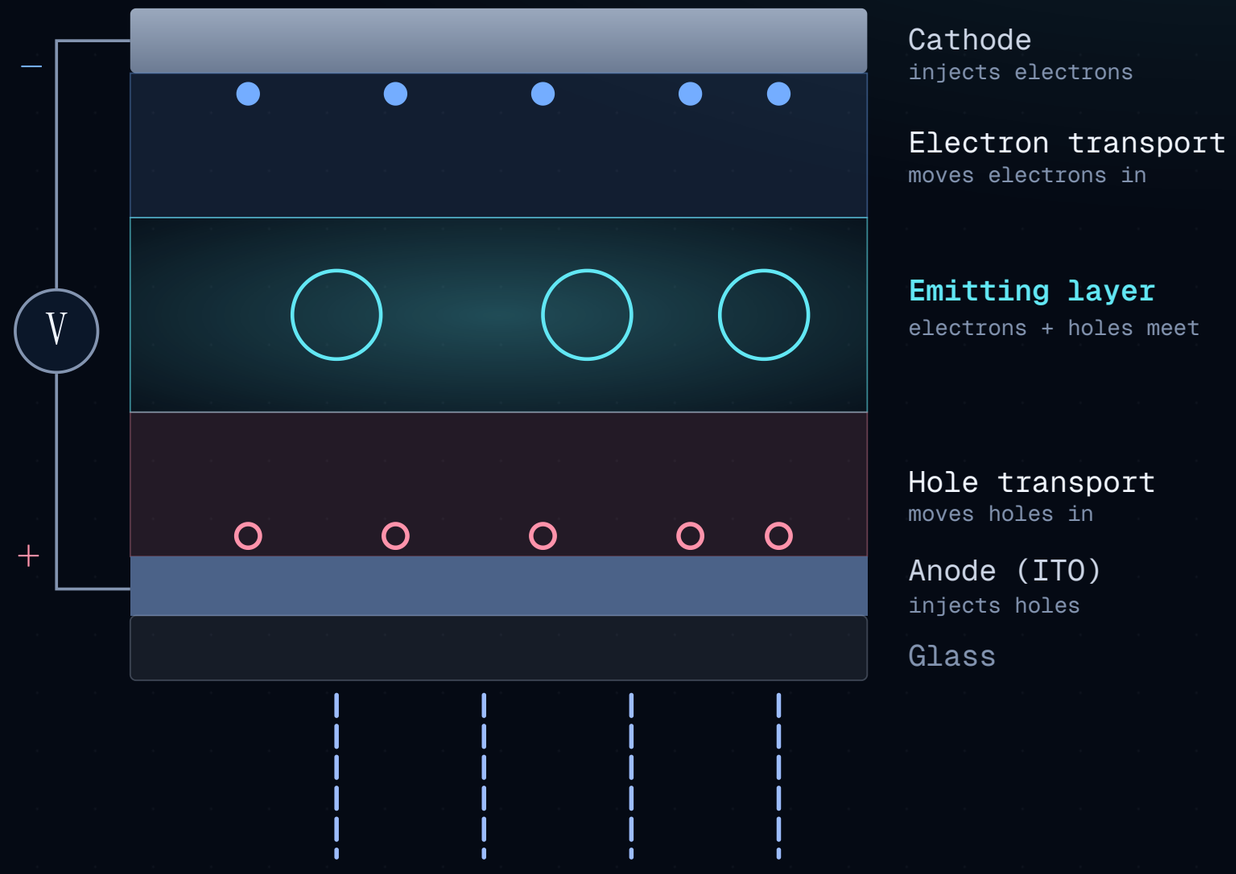
Dhana · Transferable Skills for Physicists · Assignment 2

An OLED is a *sandwich* of molecules

Voltage in. Charges move. Light out.

- **Anode** supplies holes (positive)
- **Cathode** supplies electrons
- **Organic layers** transport the charges until they meet

Organic semiconductors are built from **individual molecules**, not a continuous crystal. So we think in molecular energy levels.



Two levels matter: *HOMO* and *LUMO*

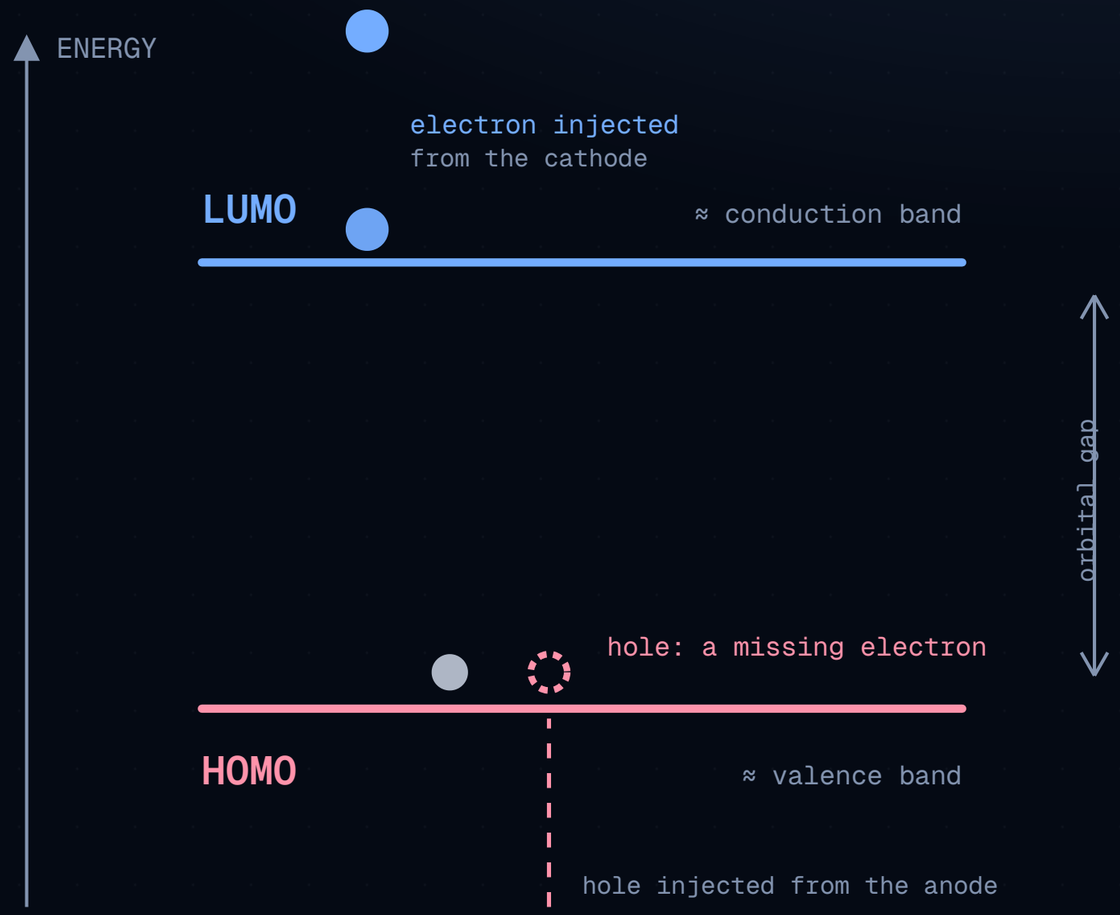
LUMO LOWEST UNOCCUPIED MOLECULAR ORBITAL

The molecular version of the **conduction band**. An injected electron sits here.

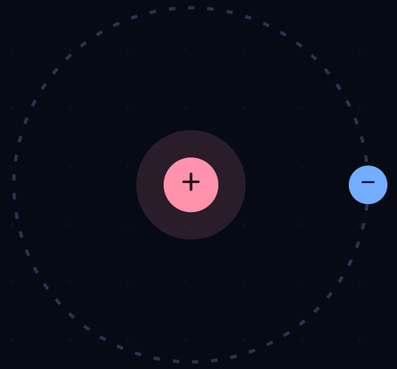
HOMO HIGHEST OCCUPIED MOLECULAR ORBITAL

The molecular version of the **valence band**. A hole is an empty state here.

Silicon: bands in a crystal. Organics: discrete levels on each molecule.



Electron meets hole: a *loose artificial atom*

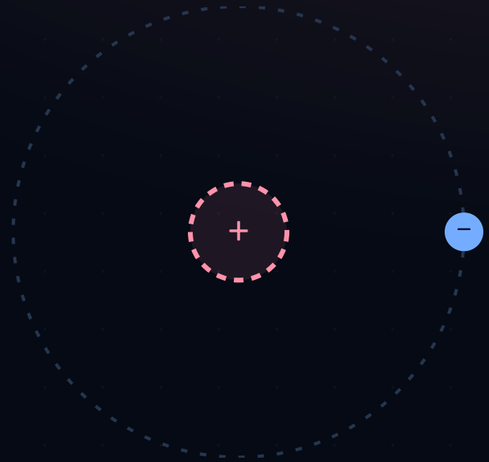


Hydrogen atom

electron bound to a proton

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0\, r}$$

≈



Exciton

electron bound to a hole

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0\varepsilon_r\, r}$$

The hole is not a particle in the molecule. It is an **absent electron** that behaves like a positive quasiparticle.

Two spin-½ objects: *one* singlet, *three* triplets

$$\tfrac{1}{2} \otimes \tfrac{1}{2} = 0 \oplus 1$$

electron spin ½ + hole spin ½ couple to total spin **S = 0** or **S = 1**.

● S = 0 · 1 state

m = 0



Singlet

$\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$

● S = 1 · 3 states

m = +1



Triplet

$|\uparrow\uparrow\rangle$

m = 0



Triplet

$\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$

m = -1



Triplet

$|\downarrow\downarrow\rangle$

1 + 3 = 4 spin states

Four doors: *1 in 4* leads to the singlet room

Throw excitons at the doors. Most end up in the triplet room.

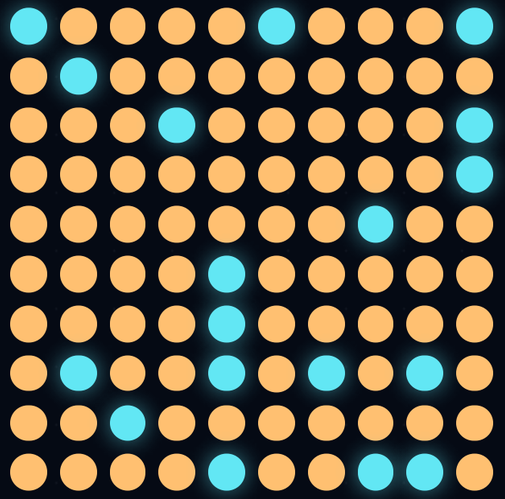
$$P(S) = \frac{1}{4}, \qquad P(T) = \frac{3}{4}$$

Throw 100 | Throw 10 000 | Reset

SINGLETS 18.0% (18)

TRIPLETS 82.0% (82)





one singlet door

three triplet doors

A sample of 100 throws. Press Throw 100 to run your own.

Two locks on one door: the *spin lock*

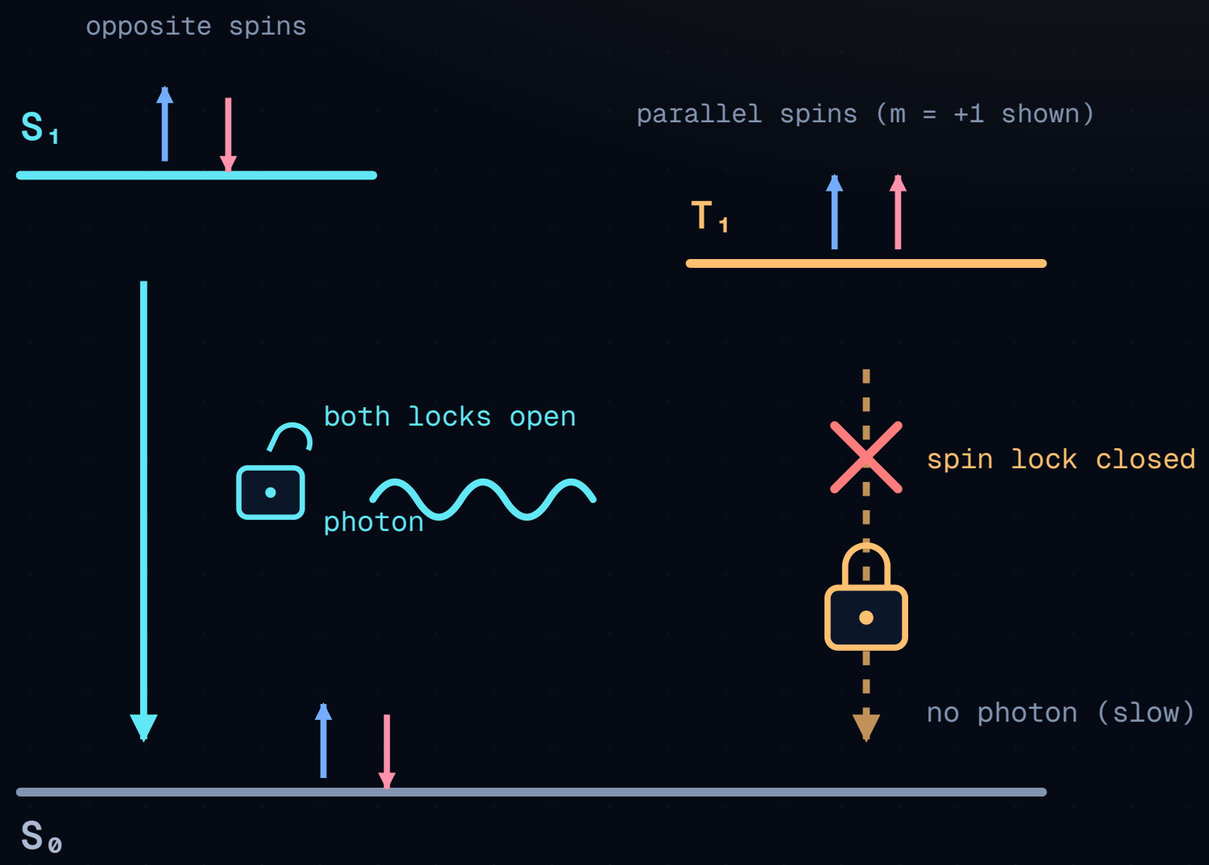
Lock 1 · energy — Is the transition energetically possible?

Lock 2 · spin — Is the spin configuration compatible?

$$\Delta S = 0 \quad \text{(electric dipole)}$$
$$\langle S_0 | \hat{\boldsymbol{\mu}} | T_1 \rangle = 0$$

Light acts on the **spatial** wavefunction. It does not directly flip an electron spin.

Forbidden ≠ impossible.
Just much less probable.



A fluorescent OLED can use only *one quarter*

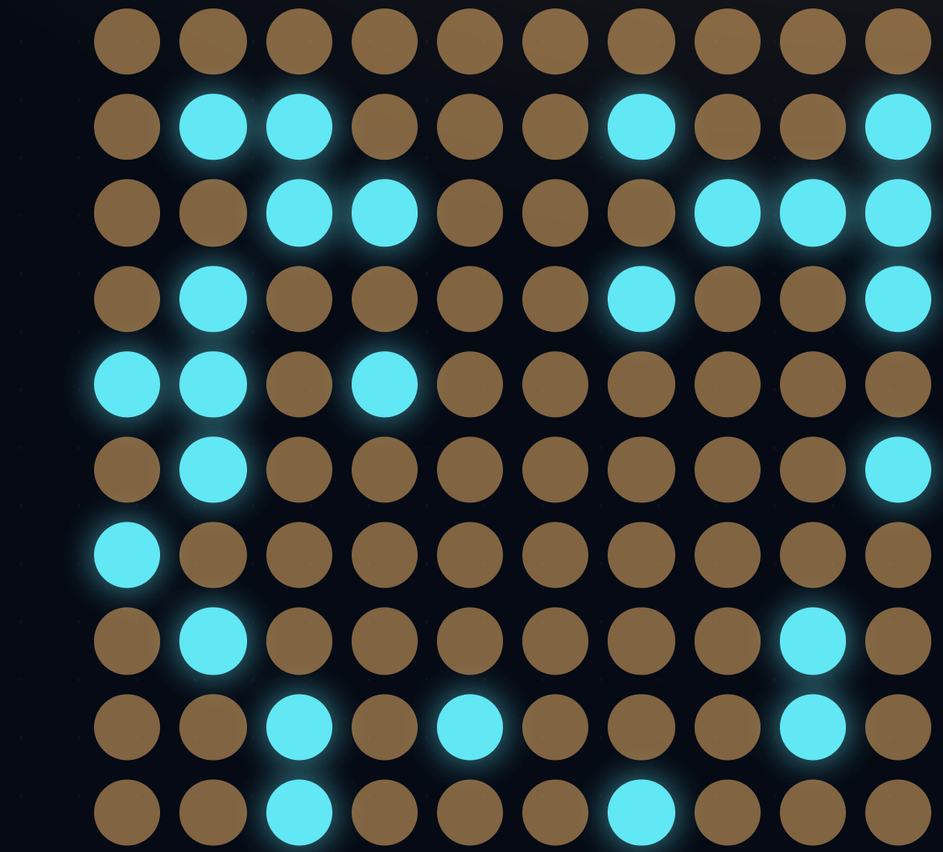
25% SINGLETS

$S_1 \rightarrow S_0$ is allowed.
They make photons.

75% TRIPLETS

$T_1 \rightarrow S_0$ is spin-forbidden.
They are stranded.

A spin-statistical problem
that TADF sets out to solve.



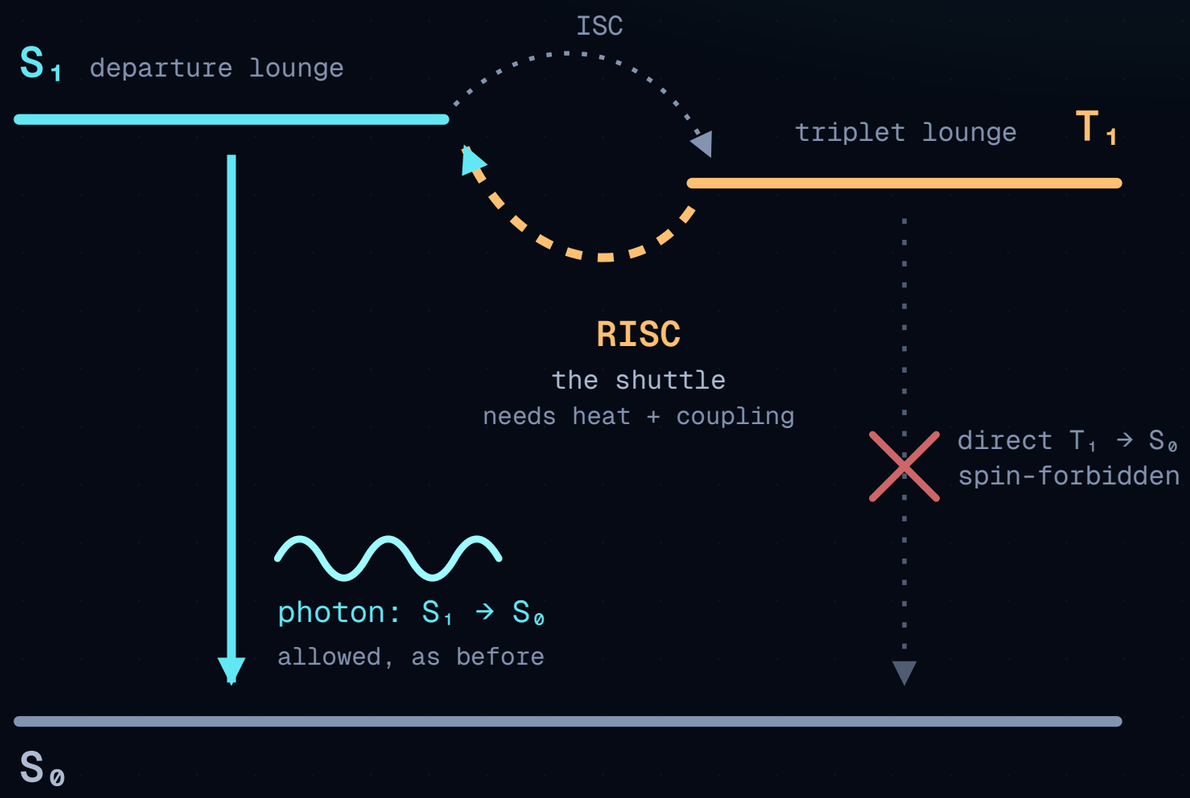
Don’t make the triplet emit. *Send it back.*

Thermally **A**ctivated **D**elayed **F**luorescence

$$T_1 \xrightarrow{\text{RISC}} S_1 \xrightarrow{h\nu} S_0$$

ISC	$S_1 \rightarrow T_1$ crosses between spin systems
RISC	$T_1 \rightarrow S_1$, the **reverse** direction

The airport: a conventional emitter leaves passengers stranded in the triplet lounge. TADF builds a **shuttle** back to the departure lounge with a direct flight.



Thermally activated: a wall or a small step?

Usually T_1 lies below S_1. The exciton must climb ΔE_{ST} using heat from its surroundings.

$$\Delta E_{ST} = E(S_1) - E(T_1)$$
$$p \sim e^{-\Delta E_{ST}/k_B T}$$

SINGLET-TRIPLET GAP **0.100 eV**

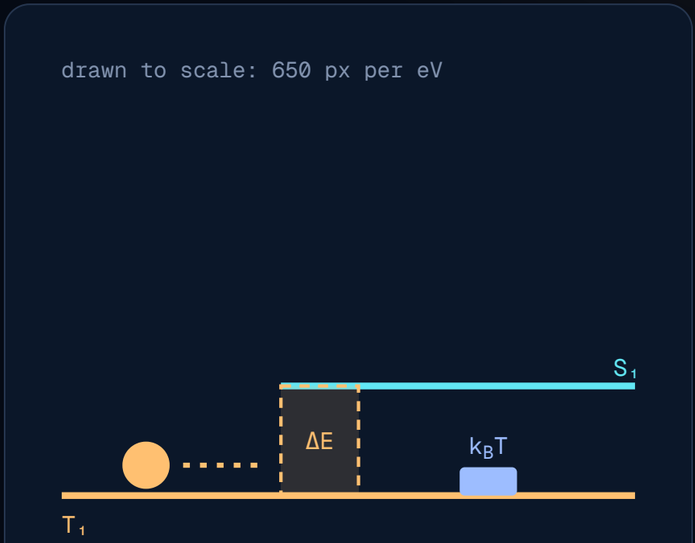
a small step — a three-metre wall

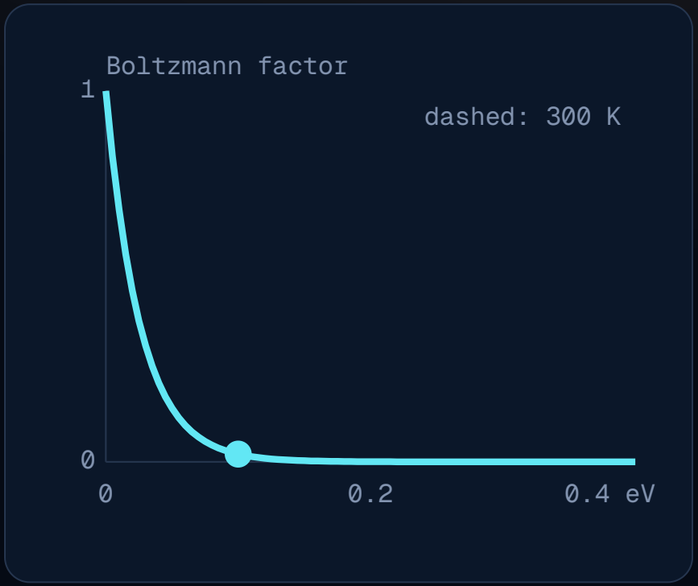
TEMPERATURE **300 K**

2.1% BOLTZMANN FACTOR

3.9 ΔE / K_BT

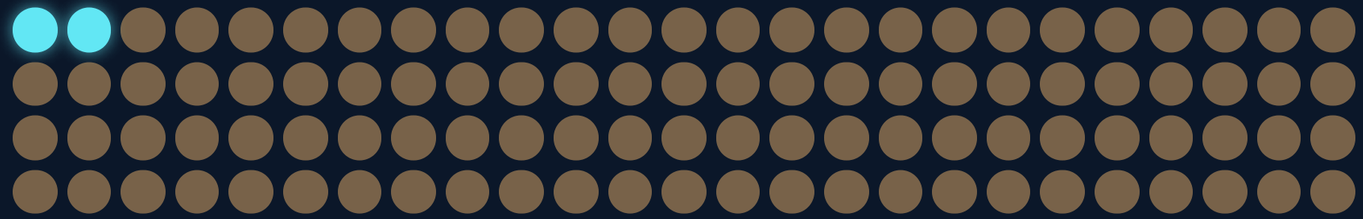
25.9 K_BT, MEV





100 THERMAL KICKS · 2 CLIMBED TO S_1

Roll 100 kicks

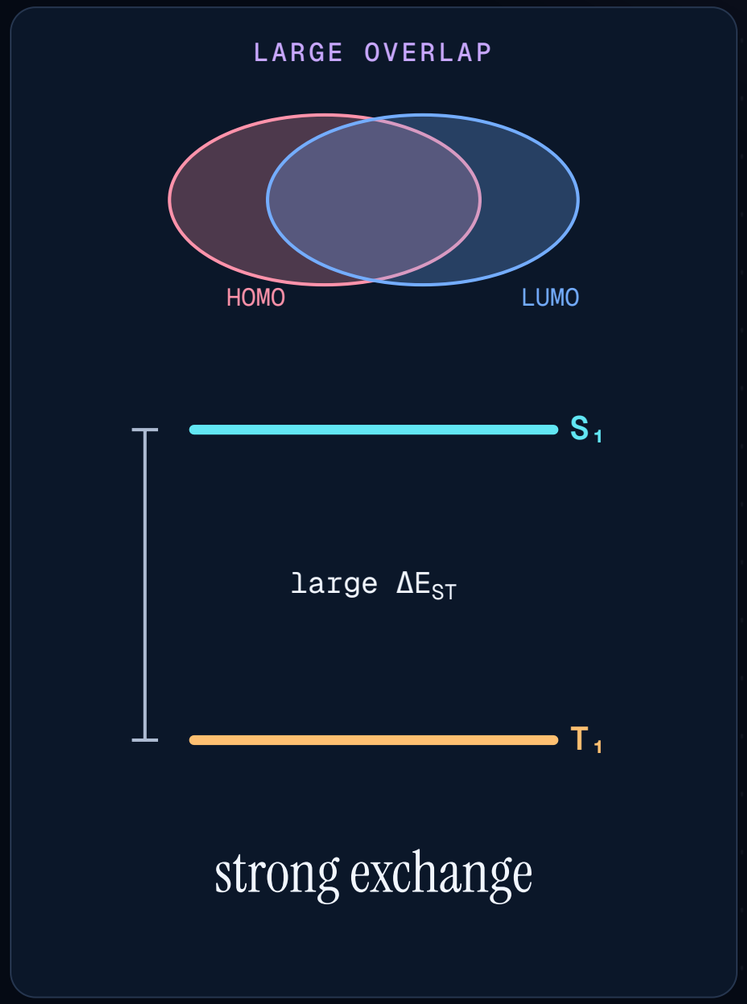


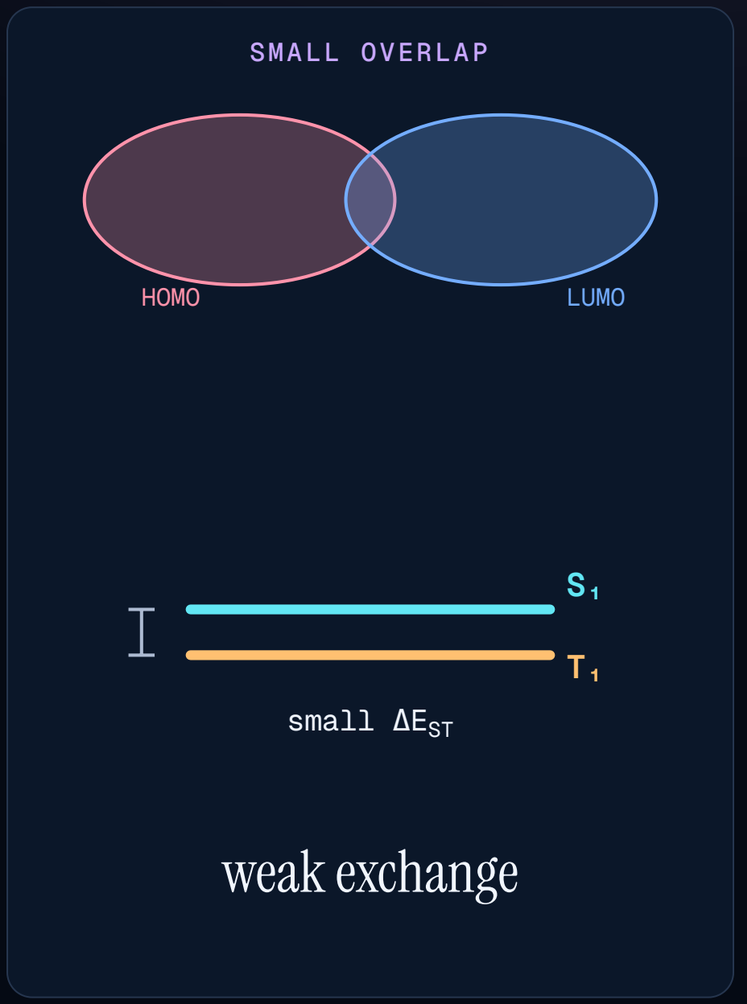
Where the gap comes from: *exchange*

Singlet and triplet have different spatial symmetry, because the full two-fermion wavefunction must stay antisymmetric.

$$\Delta E_{ST} \approx 2J$$

J is the **exchange energy**. Its size depends on how much the electron and hole wavefunctions **overlap in space**.





Pull the clouds *apart*

Put the HOMO on a donor and the LUMO on an acceptor, then twist.

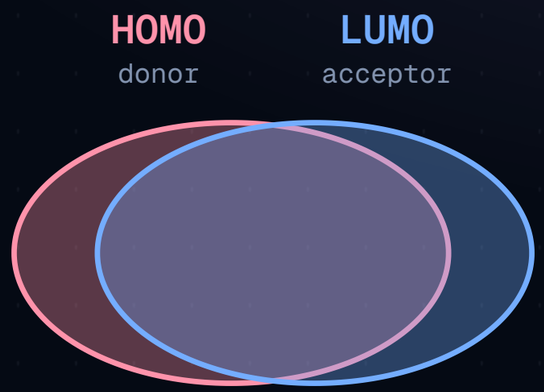
HOMO-LUMO OVERLAP **80%**

separated / twisted — on top of each other

Overlapped | Twisted | Separated

$$\Delta E_{ST} \approx 2J \quad \downarrow \text{ as overlap } \downarrow$$

Charge-transfer character: excitation moves electron density from donor towards acceptor.



SINGLET-TRIPLET GAP ΔEST **0.135 eV**

OPTICAL TRANSITION STRENGTH Kr **64 % of max**

Less overlap shrinks the gap, but it shrinks the emission too. That is the catch.

The balancing act: *not too close, not too far*

Two neighbouring houses: too close and the residents interact too strongly; too far and they cannot communicate.

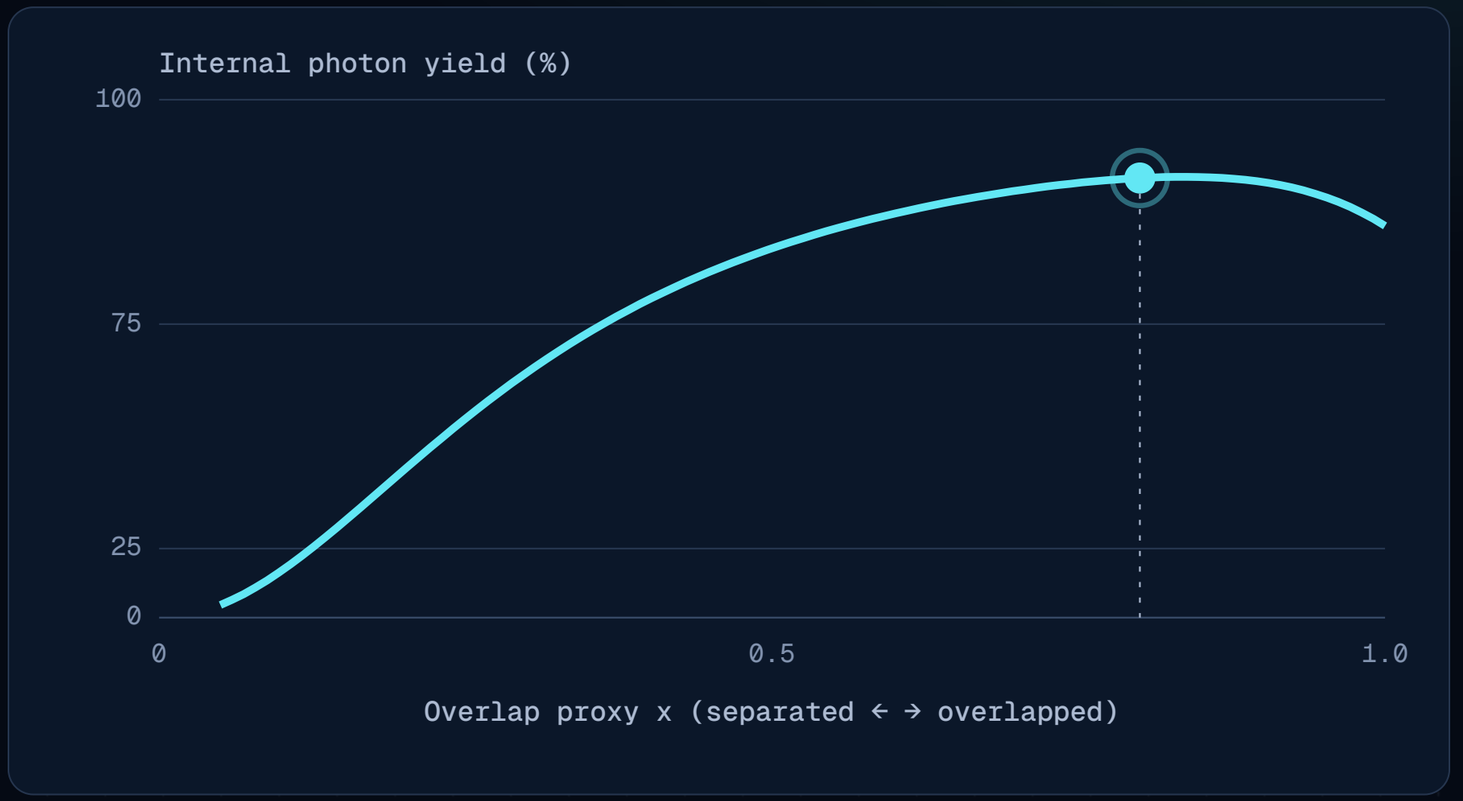
- **too close** — large J → large ΔE_{ST}
- **just right** — small gap, still bright
- **too far** — $k_r \to 0$, weak emission

$$k_r \propto \omega^3 \left|\boldsymbol{\mu}_{S_0 S_1}\right|^2$$

$$\mu \propto \int \psi_{S_0}\, \hat{\mu}\, \psi_{S_1}\, dr$$

OVERLAP PROXY X **0.80**

Low overlap | Compromise | Full overlap



84.9% INTERNAL PHOTON YIELD

0.135 $E_A \approx \Delta E_{ST}$ PROXY, EV

5.4×10^5 K_{RISC}, S^{-1}

A small gap isn't enough: *blur the boxes*

NO SPIN-ORBIT COUPLING



Pure singlet, pure triplet.
The spin lock holds.

LIGHT ATOMS · WEAK



Carbon, nitrogen, hydrogen:
only a **slight** blur.

HEAVY ATOM (IR) · STRONG



Strongly mixed: triplets can emit.
Phosphorescent OLEDs.

$$|\tilde{S}_1\rangle \approx |S_1\rangle + \lambda\,|T\rangle, \qquad \lambda \sim \frac{\langle T|\hat{H}_{SO}|S_1\rangle}{E_S - E_T}$$

$$\hat{H}_{SO} \propto \hat{\mathbf{L}} \cdot \hat{\mathbf{S}}$$

Why *delayed*? Watch the same pulse twice

Same 100 excitons, same losses.
Only RISC changes.

PROMPT

$$S_1 \rightarrow h\nu$$

DELAYED

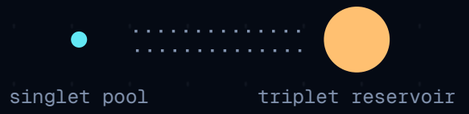
$$S_1 \rightarrow T_1 \xrightarrow{\text{wait, heat}} S_1 \rightarrow h\nu$$

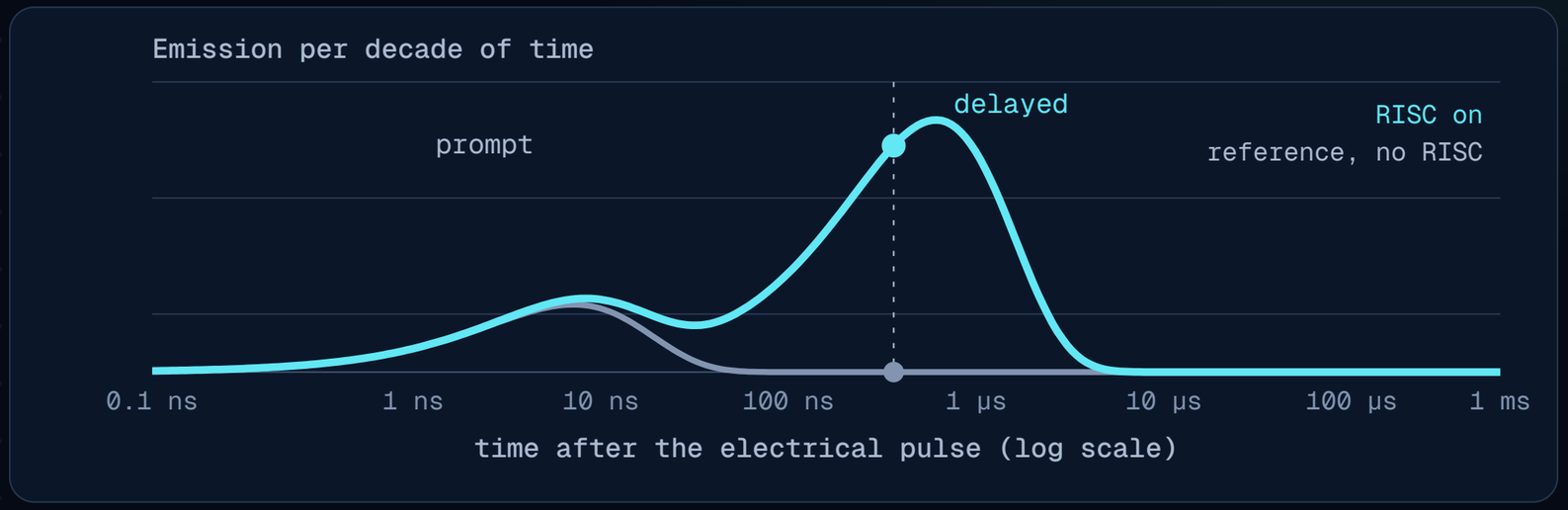
Replay pulse | Skip to end

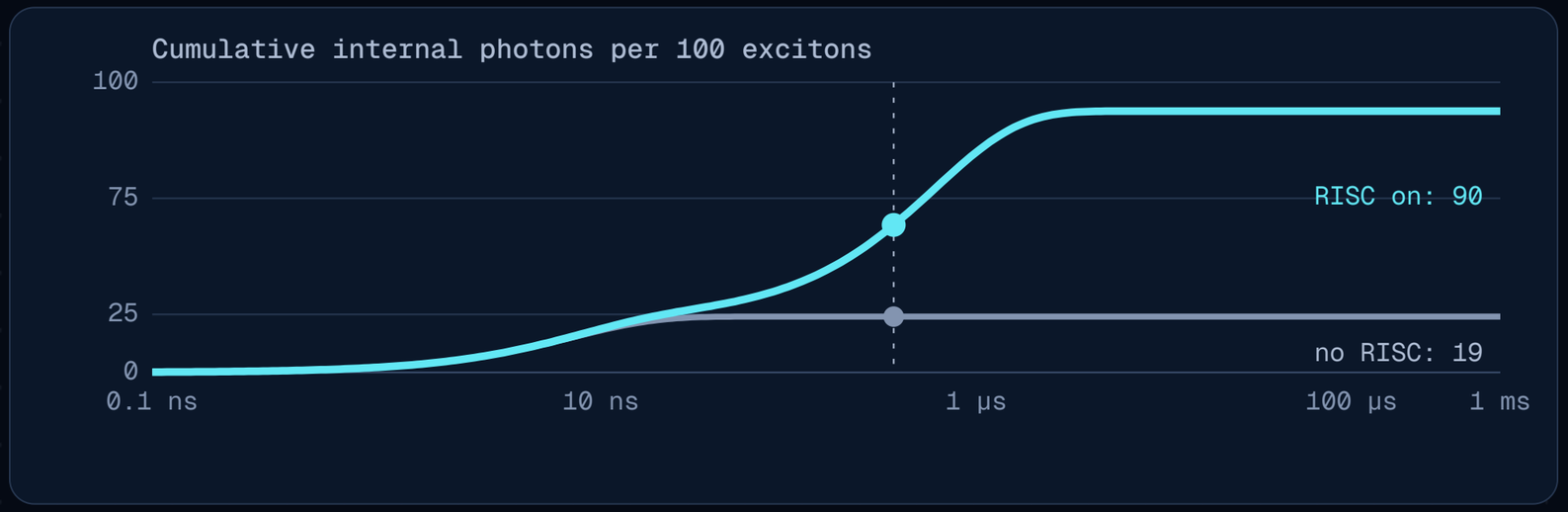
TIME AFTER PULSE **708 ns**

delayed window

19.2	50.8
reference, no RISC	RISC on, photons







Start with *one hundred* excitons

25 become singlets, 75 become triplets. A good TADF emitter **recycles** the triplets.

$$\mathrm{IQE} \approx 0.25 + 0.75\,\eta_T$$

TRIPLETS RECOVERED BY RISC (η_T) **90%**

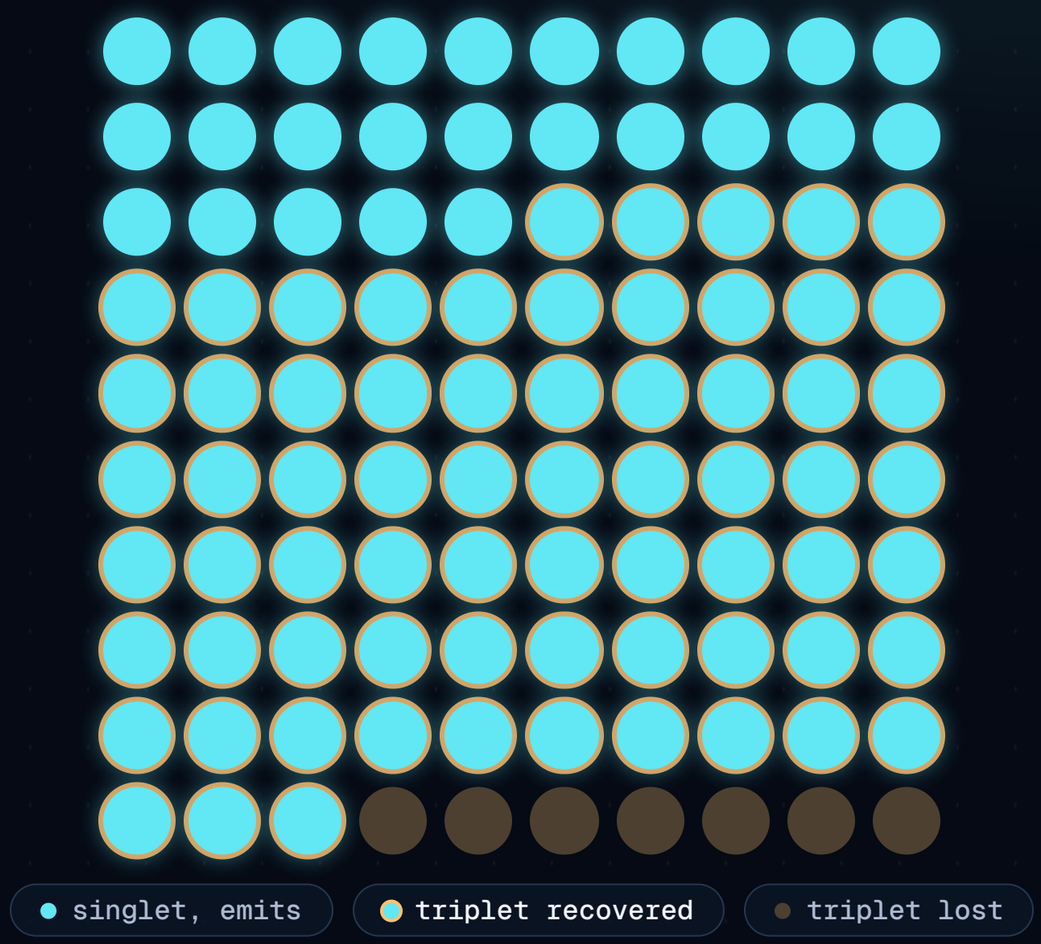
No RISC · 0%

Good TADF · 90%

Ideal · 100%

INTERNAL QUANTUM EFFICIENCY **93%**

FLUORESCENT REFERENCE 25%



93 of 100 excitons produce a photon.

Internal versus *external* efficiency

Some photons never leave: reflection, waveguiding, trapped inside the stack.

$$\mathrm{EQE} \approx \gamma\, \Phi_{\mathrm{EL}}\, \eta_{\mathrm{out}}$$
$$\mathrm{EQE} = \mathrm{IQE} \times \eta_{\mathrm{out}}$$

Fluorescent 25% | Ideal TADF 100%

LIGHT OUTCOUPLING (η_{out}) 20%



20% EXTERNAL (EQE)

100% INTERNAL (IQE)



Even near-perfect exciton use can give a much lower EQE.

The whole story, *start to finish*

01



Voltage in

Electrons from the cathode, holes from the anode, moving through organic layers.

02



Exciton

An electron and a hole bind into a loose artificial atom.

03



Spin statistics

Two spin-½ objects: 1 singlet and 3 triplets, so 25 : 75.

04



The spin lock

$S_1 \rightarrow S_0$ is allowed. $T_1 \rightarrow S_0$ breaks $\Delta S = 0$, so triplets are stranded.

05



Small gap

Separate HOMO and LUMO to cut exchange, but not so far that emission dies.

06



RISC

Heat plus spin–orbit and vibronic coupling lift T_1 back up to S_1.

07



Delayed light

$S_1 \rightarrow S_0$ fluoresces as before, now with a delayed component.

08



Nearly all excitons

Ideal limit: IQE → 100%. Outcoupling, losses and the host still matter.

Sources and *credits*

RESEARCH

[1] **Uoyama et al.**, “Highly efficient organic light-emitting diodes from delayed fluorescence”, *Nature* 492, 234 (2012). doi:10.1038/nature11687

[2] **Etherington et al.**, *Nat. Commun.* 7, 13680 (2016): local triplet states and vibronic coupling in donor–acceptor RISC. paper

[3] **Kim et al.**, *Nat. Commun.* 11, 1765 (2020), Table 2: TMCz-BO efficiency roll-off. doi:10.1038/s41467-020-15558-5

[4] **Stavrou et al.**, *Nat. Photonics* (2024): sensitizer / terminal-emitter picture. doi:10.1038/s41566-024-01395-1

COMPANY AND PRODUCT EVIDENCE

[5] **Idemitsu**, blue fluorescent OLED news release (2022). idemitsu.com

[6] **LG**, hybrid blue panel announcement (2025). lgcorp.com

[7] **Kyulux**, WiseChip 5.5-inch HF PMOLED (2023); Kyulux / SK Materials JNC agreement (2026). kyulux.com

CREDITS

img Product photo: WiseChip, via Kyulux (2023). All other graphics, equations (KaTeX) and simulations are original to this deck. Fonts: Instrument Serif, Geist, Geist Mono.

note Photon-energy conversion: $E[\text{eV}] = 1239.84 / \lambda[\text{nm}]$. Model curves are illustrative and are never presented as measured data.